

Geometry: Parallel Lines Cut by a Transversal

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DIRECTIONS

Use parallel line angle relationships to solve for x and find angle measures. Show all work.

- 1 Corresponding Angles — Solve for x :

$$\angle 1 = (3x + 20)^\circ, \quad \angle 5 = (5x - 8)^\circ$$

Answer: _____

- 2 Alternate Interior Angles — Solve for x :

$$\angle 3 = (4x + 15)^\circ, \quad \angle 6 = (7x - 12)^\circ$$

Answer: _____

- 3 Co-Interior Angles — Solve for x :

$$\angle 4 = (2x + 30)^\circ, \quad \angle 5 = (4x + 6)^\circ$$

Answer: _____

- 4 Alternate Exterior Angles — Solve for x :

$$\angle 1 = (6x - 5)^\circ, \quad \angle 8 = (4x + 25)^\circ$$

Answer: _____

- 5 Corresponding Angles — Find x :

$$\angle 2 = (2x + 10)^\circ, \quad \angle 6 = 110^\circ$$

Answer: _____

- 6 Co-Interior Angles — Find $m\angle 5$:

$$m\angle 4 = 65^\circ$$

Answer: _____

- 7 Alternate Interior Angles — Find x and $m\angle 3$:

$$\angle 3 = (5x + 12)^\circ, \quad \angle 6 = (8x - 18)^\circ$$

Answer: _____

- 8 Alternate Exterior Angles — Find x and $m\angle 1$:

$$m\angle 1 = (4x + 5)^\circ, \quad m\angle 8 = (6x - 25)^\circ$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

Corresponding, alternate interior, and alternate exterior angles are EQUAL. Co-interior (same-side interior) angles sum to 180° .

1 Corresponding Angles — Solve for x:

$$\angle 1 = (3x + 20)^\circ, \quad \angle 5 = (5x - 8)^\circ$$

$$= x = 14$$

$$3x + 20 = 5x - 8 \text{ (corresponding angles are equal)} \rightarrow 28 = 2x \rightarrow x = 14.$$

2 Alternate Interior Angles — Solve for x:

$$\angle 3 = (4x + 15)^\circ, \quad \angle 6 = (7x - 12)^\circ$$

$$= x = 9$$

$$4x + 15 = 7x - 12 \text{ (alt. interior } \angle\text{s equal)} \rightarrow 27 = 3x \rightarrow x = 9.$$

3 Co-Interior Angles — Solve for x:

$$\angle 4 = (2x + 30)^\circ, \quad \angle 5 = (4x + 6)^\circ$$

$$= x = 24$$

$$\text{Co-interior angles sum to } 180^\circ: (2x + 30) + (4x + 6) = 180 \rightarrow 6x + 36 = 180 \rightarrow x = 24.$$

4 Alternate Exterior Angles — Solve for x:

$$\angle 1 = (6x - 5)^\circ, \quad \angle 8 = (4x + 25)^\circ$$

$$= x = 15, \quad m\angle 1 = 85^\circ$$

$$6x - 5 = 4x + 25 \text{ (alt. exterior } \angle\text{s equal)} \rightarrow 2x = 30 \rightarrow x = 15.$$

5 Corresponding Angles — Find x:

$$\angle 2 = (2x + 10)^\circ, \quad \angle 6 = 110^\circ$$

$$= x = 50$$

$$2x + 10 = 110 \text{ (corresponding)} \rightarrow 2x = 100 \rightarrow x = 50.$$

6 Co-Interior Angles — Find $m\angle 5$:

$$m\angle 4 = 65^\circ$$

$$= m\angle 5 = 115^\circ$$

$$\angle 4 + \angle 5 = 180^\circ \text{ (co-interior)} \rightarrow \angle 5 = 180^\circ - 65^\circ = 115^\circ.$$

7 Alternate Interior Angles — Find x and $m\angle 3$:

$$\angle 3 = (5x + 12)^\circ, \quad \angle 6 = (8x - 18)^\circ$$

$$= x = 10, \quad m\angle 3 = 62^\circ$$

$$5x + 12 = 8x - 18 \rightarrow 30 = 3x \rightarrow x = 10. \quad m\angle 3 = 5(10) + 12 = 62^\circ.$$

8 Alternate Exterior Angles — Find x and $m\angle 1$:

$$m\angle 1 = (4x + 5)^\circ, \quad m\angle 8 = (6x - 25)^\circ$$

$$= x = 15, \quad m\angle 1 = 65^\circ$$

$$4x + 5 = 6x - 25 \text{ (alt. exterior equal)} \rightarrow 30 = 2x \rightarrow x = 15. \\ m\angle 1 = 4(15) + 5 = 65^\circ.$$

